EARTH DISTRICT ▪ SMART COMMUNITIES CHALLENGE 2026

Climate Shield

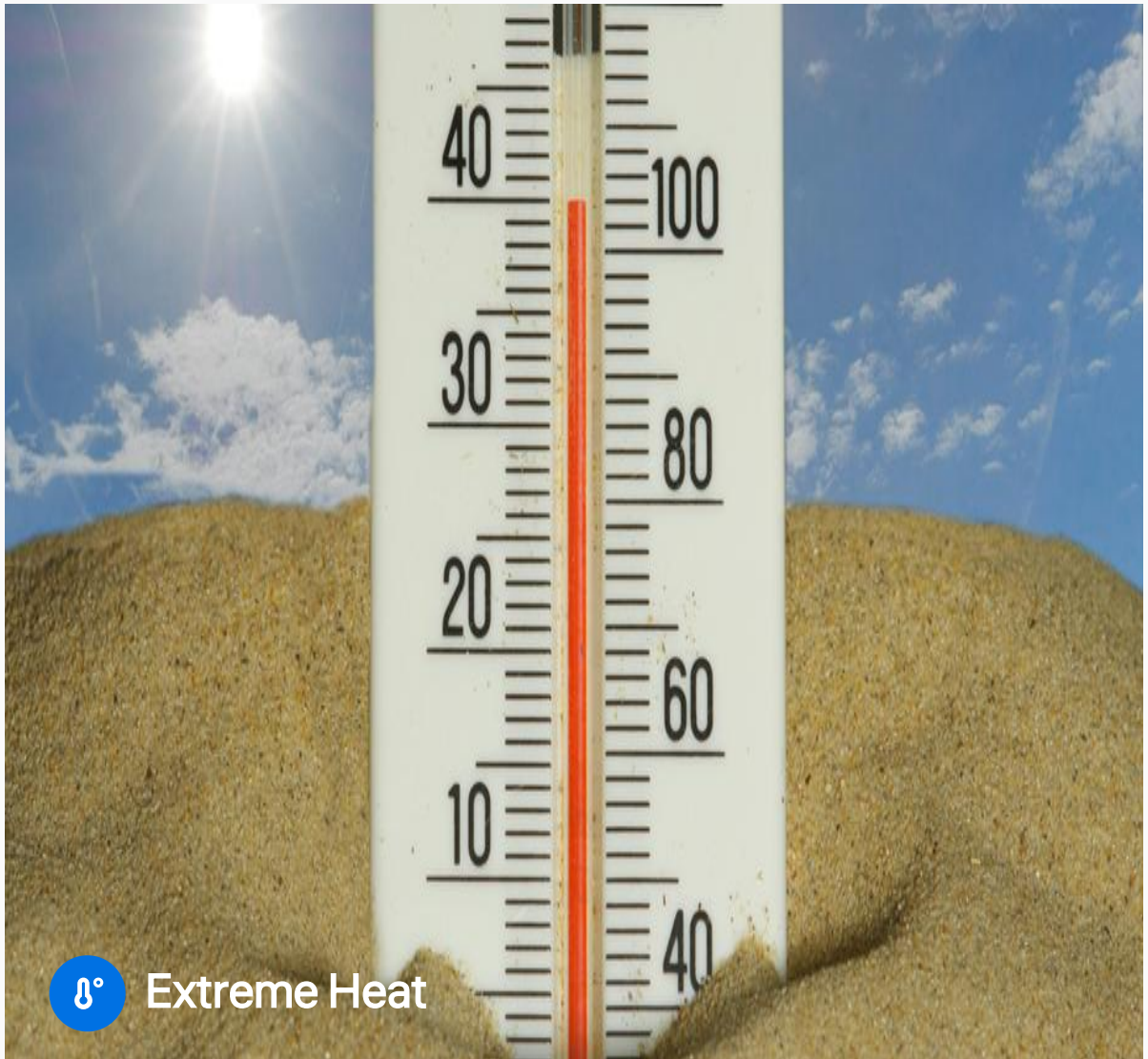
AI Climate Risk Platform

Flooding & Heat Resilience — from risk prediction to coordinated action

Sensors + city data · AI risk models · Operational playbooks · Service dispatch

Why This Matters

Flood + heat risks are rising — but response systems are still siloed



The Gap

From Alerts → Action

Today's problem isn't "no data."

It's "no coordination."

Siloed Systems

Water, energy, transit, housing, and health work off separate dashboards — if any.

Slow Awareness

Signals arrive late: crews learn about failures from calls, not sensors + models.

Alerts Don't Trigger

Residents get warnings, but not the next step: cooling sites, transport, aid, claims.

Inequitable Reach

The highest-risk neighborhoods are often the least connected to trusted channels.

Our Solution

ClimateShield turns risk signals into coordinated actions across city services.

Think: "mission control" for flood + heat response — with playbooks, not just maps.

Sense & Ingest

Sensors + city data

Predict Risk

AI models + anomaly detection

Prioritize

Assets + vulnerability + equity

Coordinate

Playbooks + dispatch + messaging

Verify

Outcomes + after-action learning

- **Earlier, neighborhood-scale** risk awareness (not just citywide alerts)
- **Clear "next best actions"** for agencies + service providers
- **Resident-facing guidance** that connects people to real services
- **Feedback loop:** after-action data improves playbooks over time

Platform Architecture

What we're building — a modular stack that cities can adopt incrementally

Engagement Layer

Resident app + SMS/IVR + partner portal + APIs

Decision Layer

Operational playbooks ▪ resource allocation ▪ incident collaboration

AI & Analytics Layer

Flood/heat risk models ▪ asset vulnerability ▪ anomaly detection ▪ scenario sims

Data Layer

Water ▪ energy ▪ transportation ▪ green infrastructure ▪ critical facilities ▪ housing & development ▪ claims/insurance signals

Built for the Real World

- Interoperable (APIs)
- Audit logs
- Data minimization
- Role-based access
- Human-in-the-loop

Designed for incremental adoption by cities of any size

User Experience

What Users See

Dashboards + playbooks + resident support — a single pane of glass

Neighborhood Risk Map

Real-time flood + heat visualization at neighborhood scale

FLOOD Pre-stage pumps near low-lying corridors

Owner: TBD Status: queued

HEAT Extend cooling center hours + notify residents

Owner: TBD Status: queued

TRANSIT Reroute buses away from flooded intersections

Owner: TBD Status: queued

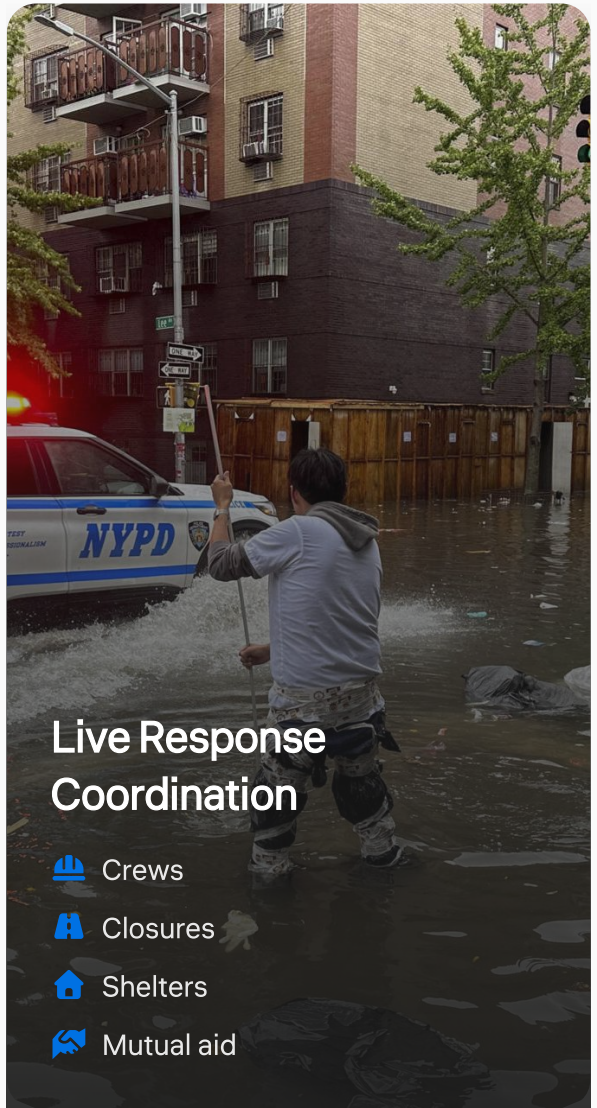
HEALTH Target wellness checks in high-risk blocks

Owner: TBD Status: queued

Operational Workflow

How It Works: Flood

From Signal → Action — Flash floods can develop quickly; coordination speed matters



Live Response Coordination

- Crews
- Closures
- Shelters
- Mutual aid

T-12h

Ingest & Monitor

Rainfall, stream gauges, 311 + social signals

T-3h

Predict & Pre-stage

Predict hotspots ▪ pre-stage crews ▪ push targeted warnings

T0

Live Incident Response

Live incident map ▪ road closures ▪ transit reroutes

T+6h

Damage Triage

Damage triage ▪ shelter capacity ▪ insurance/service handoffs

T+48h

After-Action Review

After-action review ▪ playbook update

Operational Workflow

How It Works: Heat

From Forecast → Community Support — Heat events are predictable, but impacts spike when services don't scale

Trigger "Heat Ready" Mode

Staffing, EMS surge, outreach capacity activation

Optimize Cooling Centers

Locations + hours using demand forecasting

Targeted Resident Messaging

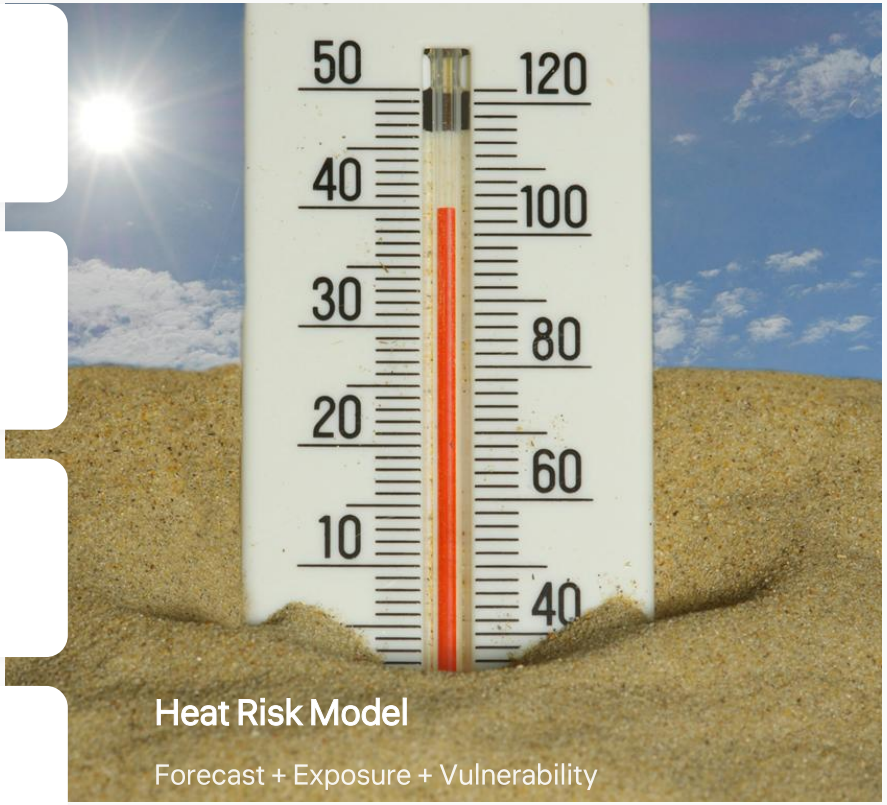
Via trusted channels (multi-lingual)

Coordinate Transport

To cooling sites for mobility-limited residents

Partner Handoffs

Nonprofits, insurers, employers, housing managers



Heat Risk Model

Forecast + Exposure + Vulnerability

Resident View

"It's hot tomorrow. Here's your nearest cooling site + a ride option."

Partnerships

Ecosystem Integrations

Where data + action come from — ClimateShield connects the sectors that already respond

- Water Systems
- Energy Grid
- Transportation
- Green Infrastructure
- Critical Facilities
- Housing & Development
- Insurance & Claims
- Community Orgs
- Health & EMS

Weather + Climate Data

Integration approach: APIs + data-sharing agreements + "minimum viable" interoperability

Key Design Choice

We don't replace existing systems. We **bridge them** with shared risk context + coordinated playbooks — so every partner can keep their tools while moving in sync.

Implementation

Pilot Plan

Start small ▪ prove value ▪ scale — A 90-day "minimum viable coordination" pilot

90-Day Pilot Scope

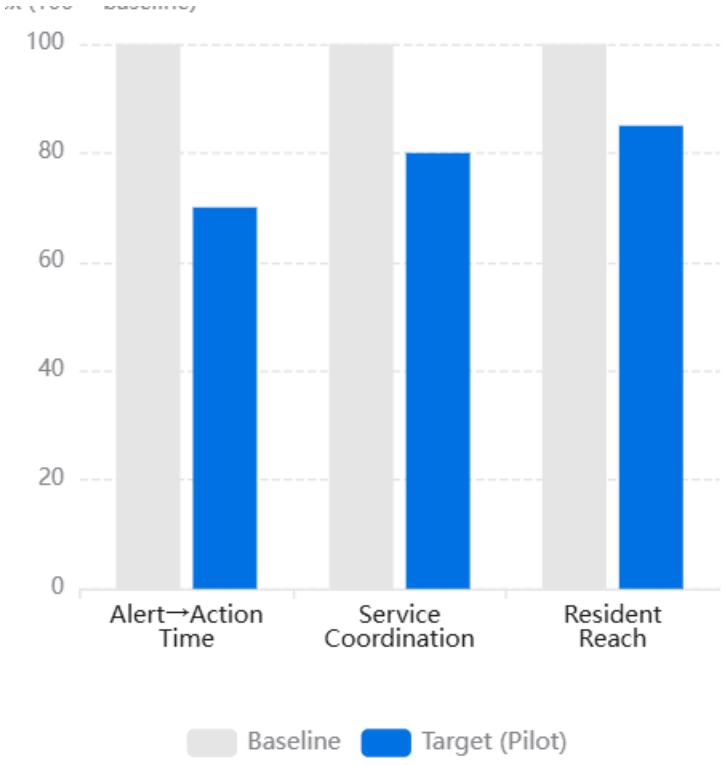
- Connect 3–5 priority datasets (rainfall + stream gauges + 311 + road closures + cooling sites)
- Deploy flood + heat risk maps at neighborhood scale
- Create 2 playbooks: Flash Flood Response and Heat Ready Mode
- Run tabletop exercise + one live drill with partner agencies
- Measure time-to-action and service coordination outcomes

Partners We Need

- City emergency management
- Transit operations
- Public health / EMS
- Community orgs
- Data owners
- Public works

Success Metrics

Illustrative targets; refined with partners during pilot



Inclusive Design

Equity by Design

Resilience that reaches people — Resilience is only real if it reaches the people most at risk

Targeted Outreach

Prioritize blocks with higher vulnerability; integrate community org distribution lists

Multi-Channel Delivery

App + SMS + IVR/call trees + partner alerts so people don't need smartphones to get help

Language + Trust

Multi-lingual templates and trusted messengers (nonprofits, faith groups, clinics)

Actionable Guidance

Not just warnings: where to go, how to get there, what services are available now

Trust & Accountability

Governance & Safety

Trust is the product — Designed for public-sector realities: privacy, procurement, and accountability

Privacy & Consent

- ✓ Data minimization by default
- ✓ Aggregate risk views where possible
- ✓ Explicit consent for resident app features
- ✓ Clear retention policies

Security & Access

- ✓ Role-based access
- ✓ Audit logs
- ✓ Encryption in transit/at rest
- ✓ Secure APIs

Human-in-the-Loop

- ✓ Models recommend actions; officials approve
- ✓ Documented assumptions + limitations
- ✓ After-action reviews improve playbooks
- ✓ Transparent incident reporting

Built for Public Sector | Procurement Ready | Accountability First

Timeline

Roadmap

Deliver value early — A pragmatic path: MVP → pilot → scale

0-3 Months
MVP

- Core ingestion + data QA
- Neighborhood risk maps (flood + heat)
- Two playbooks
- Basic dashboard + alerts

3-6 Months
Pilot

- Agency onboarding
- Tabletop + live drill
- Outcome tracking
- Refine actions + thresholds

6-12 Months
Scale

- More datasets + sectors
- Partner marketplace integrations
- Cross-city templates
- Evaluation + procurement readiness

What We Need

A pilot sponsor + data access + an agency champion to make this real

- **Pilot city partner** (90 days) to validate workflows for flood + heat
- **Access to 3–5 priority datasets** and a technical point-of-contact
- **Stakeholder introductions:** nonprofits, transit, health, utilities
- **Support for evaluation:** defining and measuring success metrics

ClimateShield

AI Climate Risk Platform for Flooding & Heat Resilience

Thank You
GitHub Repo: https://github.com/HungryBoyJun/climateshield-analysis/tree/main